
Dataset 1: doctors.csv

```
%%writefile doctors.csv
doctor_id,doctor_name,specialization,city,experience_years,consultation_f
D101,Dr. Ramesh,Cardiology,Hyderabad,12,1500
D102,Dr. Priya,Neurology,Bangalore,10,2000
D103,Dr. Anita,Dermatology,Chennai,8,1000
D104,Dr. Suresh,Orthopedics,Mumbai,15,2500
D105,Dr. Meera,Pediatrics,Delhi,6,1200
D106,Dr. Kiran,Cardiology,Hyderabad,20,3000
D107,Dr. Farhan,Neurology,Pune,5,1800
D108,Dr. Nisha,Dermatology,Kochi,9,1100
```

Dataset 2: visits.csv

```
%%writefile visits.csv
visit_id,patient_name,doctor_id,visit_date,diagnosis,bill_amount,payment_
V1001,Rahul Sharma,D101,2026-01-10,Heart Checkup,5000,Paid
V1002,Priya Reddy,D102,2026-01-10,Migraine,3500,Paid
V1003,Amit Kumar,D103,2026-01-11,Skin Allergy,2000,Pending
V1004,Sneha Patel,D104,2026-01-11,Fracture,12000,Paid
V1005,Farhan Ali,D105,2026-01-12,Fever,1500,Paid
V1006,Neha Singh,D106,2026-01-12,Heart Checkup,7000,Paid
V1007,Arjun Verma,D120,2026-01-13,Migraine,4000,Paid
V1008,Meera Nair,D103,2026-01-13,Skin Allergy,,Pending
V1009,Kiran Rao,D104,2026-01-14,Back Pain,6000,Paid
V1010,Nisha Reddy,D101,2026-01-14,Heart Checkup,5500,Paid
```

Notice:

```
doctor_id D120 invalid
bill_amount null
```

Dataset 3: hospital_config.json

```
%%writefile hospital_config.json
[
 {
   "hospital_id":"H101",
   "hospital_name":"Apollo Hospital",
   "city":"Hyderabad",
   "contact":{
      "phone":"9876500011",
      "email":"apollo@mail.com"
   },
   "services":[
      "Cardiology",
      "Neurology",
      "Dermatology"
   ]
 },
 {
   "hospital_id":"H102",
   "hospital_name":"Yashoda Hospital",
   "city":"Hyderabad",
   "contact":{
      "phone":null,
      "email":"yashoda@mail.com"
   },
   "services":[
      "Cardiology",
      "Orthopedics"
   ]
 },
 {
   "hospital_id":"H103",
   "hospital_name":"Care Hospital",
   "city":"Bangalore",
   "contact":{
      "phone":"9876500013",
      "email":null
   },
   "services":[
      "Neurology",
      "Pediatrics"
```

```
    ]
  }
]
```

CSV Exercises

```
1. Read doctors.csv.

2. Read visits.csv.

3. Display schema.

4. Count doctors.

5. Count visits.

6. Display doctors from Hyderabad.

7. Display Cardiology doctors.

8. Display doctors with experience > 10 years.

9. Display visits with bill_amount > 5000.

10. Display Pending payments.

11. Display Paid visits.

12. Find average consultation_fee by specialization.

13. Find maximum consultation_fee by specialization.

14. Count doctors by city.

15. Count doctors by specialization.

16. Find total bill_amount collected.

17. Find average bill_amount.

18. Find highest bill_amount.
```

19. Find lowest bill_amount.
20. Sort doctors by consultation_fee descending.
21. Sort visits by bill_amount descending.
22. Find null bill_amount records.
23. Fill null bill_amount with 0.
24. Create tax = bill_amount * 0.05.
25. Create final_bill = bill_amount + tax.

Join Exercises

26. Inner Join doctors and visits.
27. Left Join doctors and visits.
28. Right Join doctors and visits.
29. Full Join doctors and visits.
30. Find visits with invalid doctor_id.
31. Find doctors without visits.
32. Count visits per doctor.
33. Find revenue generated by each doctor.
34. Find highest revenue doctor.
35. Find highest revenue specialization.
36. Find average revenue by specialization.
37. Find total revenue by city.

38. Find doctor-wise visit count.

39. Find top 3 doctors by revenue.

40. Generate doctor performance report.

JSON Exercises

41. Read hospital_config.json.

42. Display schema.

43. Flatten contact.phone.

44. Flatten contact.email.

45. Find hospitals with null phone.

46. Find hospitals with null email.

47. Fill missing phone values.

48. Fill missing email values.

49. Display hospital_name and city.

50. Display hospital_name and phone.

51. Explode services array.

52. Display one row per service.

53. Count hospitals by city.

54. Count hospitals by service.

55. Find hospitals providing Cardiology.

56. Find hospitals providing Neurology.

57. Find hospitals providing Orthopedics.

58. Find hospitals providing Pediatrics.
59. Save flattened JSON as Parquet.
60. Save flattened JSON as CSV.

Window Function Exercises

61. Rank doctors by revenue.
62. Dense rank doctors by revenue.
63. Row number by revenue.
64. Find top doctor.
65. Find top 3 doctors.
66. Find top doctor by specialization.
67. Find top 2 doctors per specialization.
68. Create running revenue.
69. Use lag() on revenue.
70. Use lead() on revenue.
71. Compare doctor revenue with previous doctor.
72. Compare doctor revenue with next doctor.
73. Find highest revenue doctor in each city.
74. Find lowest revenue doctor in each city.
75. Generate doctor leaderboard.

Spark SQL Exercises

76. Create temp view doctors.

77. Create temp view visits.

78. Create temp view hospitals.

79. Display all doctors using SQL.

80. Count doctors by specialization.

81. Count doctors by city.

82. Find revenue by doctor.

83. Find revenue by specialization.

84. Find top 5 doctors by revenue.

85. Find pending payment visits.

86. Find hospitals providing Cardiology.

87. Find hospitals providing Neurology.

88. Find hospitals with missing contact details.

89. Find average consultation fee.

90. Generate doctor performance report using SQL.

ETL Exercises

91. Read CSV and JSON.

92. Clean null values.

93. Flatten JSON.

94. Join doctor and visit datasets.

95. Create revenue calculations.

96. Create doctor ranking.

97. Create specialization summary.

98. Save Silver layer as Parquet.

99. Create Gold reports.

100. Generate Hospital Operations Analytics Dashboard dataset.